

Artificial Intelligence 2 Solutions

2024

Artificial Intelligence 2

Exam paper

Question 1 (2021 AI2 Exam question)

As a data scientist in a telecommunication company, your task is to analyse a customer dataset to predict whether a customer will terminate his/her contract. The dataset consists of around 8000 customer records, each consisting of one binary dependent variable Y , indicating whether the customer terminates the contract ($Y = 1$) or not ($Y = 0$), and 19 independent variables, which include the customer's information, e.g., age, subscription plan, extra data plan, etc., and the consumer behaviour such as average numbers of calls and hours per week. Since your boss needs some actionable insights to retain customers, you decided to use interpretable machine learning methods. Design your interpretable machine learning method by answering the following questions:

- (a) You have implemented a feature selection algorithm based on mutual information to select the most informative features from the 19 independent variables. To validate the implementation of your mutual information calculation function, you use a small subset of the data to calculate mutual information manually. You select one independent variable 'subscription plan', denoted as S , which takes two values, $S \in \{1, 2\}$. Please use the following Probability Mass Function table

$p(S, Y)$	$S = 1$	$S = 2$
$Y = 0$	$\frac{2}{12}$	$\frac{5}{12}$
$Y = 1$	$\frac{2}{12}$	$\frac{3}{12}$

to calculate

- Entropies $H(S)$ and $H(Y)$
- Conditional entropies $H(S|Y)$ and $H(Y|S)$
- Joint entropy $H(S, Y)$
- Mutual information $I(S; Y)$

Show all your working. Discuss what mutual information means and whether this feature will be selected or not.

- (b) After applying your algorithm you selected two variables: 1) extra data plan E , which is a binary random variable that indicates whether the customer subscribes to the extra data plan ($E = 1$) or not ($E = 0$); and 2) averaged hours used per week H , which is a continuous random variable. You then built a logistic regression model

to classify customers into 'low risk' or 'high risk' of terminating the contract. The fitted model is

$$\log \left(\frac{p}{1-p} \right) = -0.77 + 0.23H - 1.18E$$

- Given a customer who has the extra data plan ($E = 1$) and spent on average 0.5 hours per week, calculate the odds and the probability the customer will terminate the contract ($Y = 1$). **(4 marks)**
- Using this fitted model, explain to your boss what actions should be taken to retain customers. **(10 marks)**

0 marks for question not valid - all questions must have the same number of marks

Model answer / LOs / Creativity:

(a)

$p(S, Y)$	$S = 1$	$S = 2$	$p(Y)$
$Y = 0$	$\frac{2}{12}$	$\frac{5}{12}$	$\frac{7}{12}$
$Y = 1$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{5}{12}$
$p(S)$	$\frac{4}{12}$	$\frac{8}{12}$	1

$$H(S) = - \sum_i^n p(s_i) \log_2 p(s_i) = - \left(\frac{4}{12} \log \frac{4}{12} + \frac{8}{12} \log \frac{8}{12} \right) = 0.92 \text{ bits}$$

$$H(Y) = - \sum_i^n p(y_i) \log_2 p(y_i) = - \left(\frac{7}{12} \log \frac{7}{12} + \frac{5}{12} \log \frac{5}{12} \right) = 0.97 \text{ bits}$$

$$H(S, Y) = - \sum_{s_i \in R_S} \sum_{y_j \in R_Y} p(s_i, y_j) \log p(s_i, y_j) = 1.89 \text{ bits}$$

$$H(S|Y) = H(S, Y) - H(Y) = 1.89 - 0.97 = 0.92 \text{ bits}$$

$$H(S|X) = H(S, Y) - H(S) = 1.89 - 0.92 = 0.97 \text{ bits}$$

$$I(S; Y) = H(S) - H(S|Y) = 0.92 - 0.92 = 0 \text{ bits}$$

Since mutual information measures the information that two random variables S and Y share. In other words, it measures how much knowing one of these variables reduces uncertainty about the other, the value of 0 means this feature is not useful at all.

(b) **First question:**

- Odds:

$$\begin{aligned} odds &= \frac{p}{1-p} = \exp(-0.77 + 0.23H - 1.18E) = \exp(-0.77 + 0.115 - 1.18) \\ &= 0.1596 \end{aligned} \quad (1)$$

- Probability:

$$P(Y = 1) = 0.137 \quad (2)$$

Second question: The student should extend the logistic model with one independent variable as learned in the lecture to two independent variables. The student should be able to investigate the effect of each variable by deriving the odd ratios by fixing the value of the other variable. The following derivations would be expected but other ways of quantifying the effects are acceptable.

$$OR_E = \frac{\text{odds when } E = 1}{\text{odds when } E = 0} = \frac{\exp(-0.77 + 0.23 - 1.18)}{\exp(-0.77 + 0.23)} = \exp(-1.18) \approx 0.31$$

$$OR_H = \frac{\text{odds when } H = h + \Delta}{\text{odds when } H = h} = \frac{\exp(-0.77 + 0.23(H + \Delta) - 1.18E)}{\exp(-0.77 + 0.23H - 1.18E)} = (\exp(0.23))^\Delta \approx 1.25^\Delta$$

The following key points should be mentioned:

- If a customer add the extra data plan, the odds of terminating the contract will increase is 0.3, which means that odds the customer terminating the contract will decrease by a factor of 3.
- If a customer increase the average time by one hour, the odds of terminating the contract increase by a factor of 1.25.

From the analysis, we can suggest to the boss that, the more hours the customers spent, the more likely the customers will terminate, which means the company should improve its telecommunication service/price. However, by simply persuade them to subscribe to the extra data plan, they are more likely to stay.

Question 2 (2022 AI2 Exam question)

As a machine learning expert for an AI cyber security company, your task is to design an automated network intrusion detection system. You have collected a large number of records of network activities. Each record includes the log information about network activity, such as protocol types, duration, number of failed logins, which are random variables, denoted as $\mathbf{X} = [X_1, X_2, \dots, X_n]^T$. Each record also includes a binary random variable Y called label that was labelled by cyber security experts as intrusions ($Y = 1$) or normal connections ($Y = 0$).

(a) Consider feature selection based on mutual information to reduce the number of independent variables.

(i) Explain to your colleague, who knows nothing about information theory, the concept of mutual information. **[2 marks]**

(ii) Explain the loop, i.e., lines 4-7 of the pseudocode in Table 1. Note $I(Y; X_i)$ is the mutual information between Y and X_i .

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1: Initialisation: Set  $F \leftarrow \mathbf{X}$  and  $S \leftarrow \emptyset$ 
2:    $f_{\max} = \operatorname{argmax}_{X_i \in \mathbf{X}} I(Y; X_i)$ 
3:   Set  $F \leftarrow F \setminus \{f_{\max}\}$  and  $S \leftarrow f_{\max}$ 
4:   Repeat until  $|S| = K$ :
5:      $f_{\max} = \operatorname{argmax}_{X_i \in F} I(Y; X_i) - \beta \sum_{X_s \in S} I(X_s; X_i)$ 
6:     Set  $F \leftarrow F \setminus \{f_{\max}\}$  and  $S \leftarrow f_{\max} \cup S$ 
7:   End
8: End

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Table 1: Pseudocode of Mutual Information based Feature Selection Algorithm.

[3 marks]

5 marks for question not valid - all questions must have the same number of marks

Model answer / LOs / Creativity:

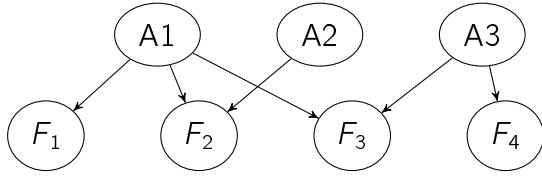
(a) Mutual Information Feature selection

(i) Basic motivation behind mutual information is to measure the information that two random variables X and Y share. In other words, it measures how much knowing one of these variables reduces uncertainty about the other.

(ii) The student should explain the two lines similar to the following: "The two lines in the loop are used to select K features. In this loop, we also find the feature $f_{\max}$ which achieves the maximum mutual information I among all the remaining independent variables in set F . However, because some features highly correlated with each other, selecting them will increase the number of features but does not improve the prediction. Therefore, we need to make sure there must be minimal redundancy between the candidate feature X_i and the set of selected features S . That's exactly the second term of the equation ($f_{\max}$) You then add this feature into S and then subtract it from set F and repeat until we got K features".

(b) Bayesian Networks

(i) The following draw is acceptable:



(ii) The joint probability distribution represented by the Bayesian network:

$$\begin{aligned}
 &P(A_1, A_2, A_3, F_1, F_2, F_3, F_4) \\
 &= P(A_1)P(A_2)P(A_3)P(F_1|A_1)P(F_2|A_1, A_2)P(F_3|A_1, A_3)P(F_4|A_3) \quad (3)
 \end{aligned}$$

(iii) Total number of parameter is 15, but the student should show the following working:

Conditional Probability	number of parameters
$P(A_1)$	1
$P(A_2)$	1
$P(A_3)$	1
$P(F_1 A_1)$	2
$P(F_2 A_1, A_2)$	4
$P(F_3 A_1, A_3)$	4
$P(F_4 A_3)$	2

(iv) With $F_2 = 1$, observing $F_4 = 1$ still gives us information only about A_3 . If we observe $F_3 = 1$ instead of F_2 , then observing $F_4 = 1$ will give us information about A_1 and A_3 due to competing causes (aka. explaining away).